

AgriConnect - USSD Gateway & OAuth2 Documentation

Overview

A webhook endpoint for USSD gateway integrations (e.g., Africa's Talking) has been added. It is secured using OAuth2. The gateway must authenticate as a Client Application and forward the user's phone number and input.

Endpoint Details

URL: POST /api/ussd/

Authentication: OAuth2 Access Token required (Authorization: Bearer <token>).

Request Body (JSON or Form URL-Encoded):

```
{
  "phoneNumber": "0240001111",
  "text": "1*2",
  "sessionId": "12345",
  "serviceCode": "*920*44#"
}
```

Response: Returns plain text USSD response strings prefixed with CON (Continue) or END (End transaction).

OAuth2 Setup Instructions

Step 1: Create OAuth Application

1. Log into the Django Admin portal (/admin/).
2. Add a new application under **Django OAuth Toolkit**.
3. Set **Client Type** to *Confidential* and **Authorization Grant Type** to *Client credentials*.
4. Give the application a descriptive name (e.g., "USSD Gateway").
5. Save the application.
6. Note down the generated *Client ID* and *Client Secret*.

Step 2: Obtain Access Token

The external gateway or emulator must request an access token by sending a POST request to the token endpoint:

```
POST /o/token/
Content-Type: application/x-www-form-urlencoded

grant_type=client_credentials&
client_id=<YOUR_CLIENT_ID>&
client_secret=<YOUR_CLIENT_SECRET>
```

This will return a JSON response containing an `access_token`. This token must be included in the Authorization header (Bearer <token>) of all subsequent requests to the /api/ussd/ endpoint.

Step 3: Using the USSD Emulator

If you are using the provided local USSD emulator, you need to configure it with the credentials you just created.

1. Open the emulator script (e.g., `ussd_emulator.py`).
2. Locate the configuration variables (`CLIENT_ID`, `CLIENT_SECRET`).
3. Paste your *Client ID* and *Client Secret* into the respective variables.
4. Start the emulator. It is designed to automatically handle fetching and refreshing the OAuth2 access token before dispatching USSD session data to the Django backend.